

# S-Duality = Langlands Duality = Koszul Duality

## for Quantum Vertex Chiral Groups

Research note for Volume III

### Abstract

We develop the triple identification

$$\text{S-duality} = \text{Langlands duality} = \text{Koszul duality}$$

in the framework of quantum vertex chiral groups. Starting from  $\mathcal{N} = 4$  super Yang–Mills in four dimensions, the Kapustin–Witten twist and compactification on a curve  $C$  produce the geometric Langlands correspondence. The two sides of geometric Langlands —  $\text{D-mod}(\text{Bun}_G(C))$  and  $\text{QCoh}(\text{LocSys}_{G^L}(C))$  — are controlled respectively by the affine Kac–Moody vertex algebra  $V_k(\mathfrak{g})$  at the critical level  $k = -h^\vee$  and by the Langlands dual group  $G^L$ . In our framework,  $V_k(\mathfrak{g})$  determines a quantum vertex chiral group  $G(C, \mathfrak{g})$ . We argue that its Koszul dual is  $G(C, \mathfrak{g}^L)$  — that is, Langlands duality *is* Koszul duality of quantum vertex chiral groups. The evidence comes from Volume I, where the Koszul dual of  $V_k(\mathfrak{g})$  involves  $V_{k'}(\mathfrak{g})$  at the dual level  $k' = -k - 2h^\vee$ , which at the critical level recovers the Feigin–Frenkel center  $\mathfrak{z}(\hat{\mathfrak{g}}) \simeq \text{Fun Op}_{G^L}(C)$ . The Hitchin moduli space  $\mathcal{M}_H(C, G)$  provides the Calabi–Yau geometry: it is hyper-Kähler (hence non-compact CY), and S-duality exchanges  $\mathcal{M}_H(C, G)$  with  $\mathcal{M}_H(C, G^L)$ , exchanging the root data of the corresponding quantum vertex chiral groups.

## Contents

### 1 S-duality in 4d $\mathcal{N} = 4$ super Yang–Mills

#### 1.1 The coupling constant and Montonen–Olive duality

Let  $G$  be a compact simple Lie group. The  $\mathcal{N} = 4$  super Yang–Mills theory with gauge group  $G$  on a four-manifold  $M^4$  depends on a complexified coupling constant

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2} \in \mathbb{H},$$

where  $\theta$  is the theta-angle and  $g$  is the gauge coupling. The Montonen–Olive conjecture (established at the level of the BPS spectrum by Sen, and at the level of partition functions by Vafa–Witten) asserts:

**Conjecture 1.1** (Montonen–Olive / S-duality). *The  $\mathcal{N} = 4$  SYM theory with gauge group  $G$  at coupling  $\tau$  is equivalent to the  $\mathcal{N} = 4$  SYM theory with gauge group  $G^L$  (the Langlands dual group) at coupling  $-1/n_{\mathfrak{g}}\tau$ , where  $n_{\mathfrak{g}} = 1$  for simply-laced  $\mathfrak{g}$  and  $n_{\mathfrak{g}} = 2, 3$  for  $B_n/C_n$  and  $G_2$  respectively.*

In the simply-laced case, the S-duality group is  $\mathrm{SL}_2(\mathbb{Z})$  acting on  $\tau$  by Möbius transformations, generated by

$$T: \tau \mapsto \tau + 1, \quad S: \tau \mapsto -1/\tau.$$

The  $T$ -transformation shifts the theta-angle and is perturbatively manifest. The  $S$ -transformation is deeply non-perturbative: it exchanges strong and weak coupling and simultaneously replaces  $G$  by  $G^L$ .

*Remark 1.2* (The Langlands dual group). The Langlands dual  $G^L$  is defined by interchanging the root and coroot lattices:

$$\Lambda_{\mathrm{root}}(G^L) = \Lambda_{\mathrm{coroot}}(G), \quad \Lambda_{\mathrm{coroot}}(G^L) = \Lambda_{\mathrm{root}}(G).$$

For simply-laced groups,  $G^L \simeq G/Z(G)$  or  $\tilde{G}$ , depending on whether  $G$  is adjoint or simply-connected. The key non-simply-laced examples are  $B_n^L = C_n$ ,  $C_n^L = B_n$ , and  $G_2^L = G_2$  (but with long and short roots exchanged).

## 1.2 The Kapustin–Witten twist

The  $\mathcal{N} = 4$  theory admits a one-parameter family of topological twists, parametrized by  $t \in \mathbb{CP}^1$ , constructed by Kapustin and Witten. The twist combines two of the supercharges  $Q_\ell$  and  $Q_r$  into a single nilpotent scalar supercharge

$$Q_t = Q_\ell + t Q_r, \quad Q_t^2 = 0.$$

- At  $t = 0$ : the **A-twist**, producing a topological field theory of Donaldson type (the “holomorphic” twist).
- At  $t = 1$ : the **GL-twist** (geometric Langlands twist), producing a balanced topological-holomorphic theory.
- At  $t = i$ : the **B-twist**, producing the Kapustin–Witten theory whose equations are the Hitchin equations.

Under S-duality, the twist parameter transforms as  $t \mapsto t/|t|^2$  (up to the action of the R-symmetry), so the A-twist at coupling  $\tau$  for gauge group  $G$  is exchanged with the B-twist at coupling  $-1/\tau$  for  $G^L$ .

**Proposition 1.3** (Kapustin–Witten). *The GL-twist of  $\mathcal{N} = 4$  SYM on  $M^4 = \Sigma \times C$ , where  $\Sigma$  is a surface and  $C$  is a smooth projective curve, produces a 4d topological field theory whose reduction on  $C$  gives a 2d  $\sigma$ -model into the Hitchin moduli space  $\mathcal{M}_H(C, G)$ . Under S-duality, the A-model on  $\mathcal{M}_H(C, G)$  is exchanged with the B-model on  $\mathcal{M}_H(C, G^L)$ .*

This is the physics origin of the geometric Langlands correspondence:

$$\text{A-model on } \mathcal{M}_H(C, G) \xleftarrow{\text{S-duality}} \text{B-model on } \mathcal{M}_H(C, G^L)$$

The A-model produces a category of branes — the Fukaya category of  $\mathcal{M}_H(C, G)$  — while the B-model produces the derived category of coherent sheaves. Homological mirror symmetry for the Hitchin system then gives the categorical equivalence underlying geometric Langlands.

## 2 Geometric Langlands

### 2.1 The correspondence

Let  $C$  be a smooth projective curve over  $\mathbb{C}$  of genus  $g \geq 2$ ,  $G$  a reductive algebraic group, and  $G^L$  its Langlands dual. The geometric Langlands conjecture (Beilinson–Drinfeld, with the categorical formulation due to Arinkin–Gaitsgory, recently proved by Gaitsgory et al.) asserts:

**Theorem 2.1** (Geometric Langlands — Gaitsgory et al.). *There is an equivalence of DG categories*

$$\mathbb{L}: \mathrm{D}\text{-mod}(\mathrm{Bun}_G(C)) \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{G^L}(C))$$

where:

- $\mathrm{Bun}_G(C)$  is the moduli stack of principal  $G$ -bundles on  $C$ ;
- $\mathrm{D}\text{-mod}(\mathrm{Bun}_G(C))$  is the DG category of  $D$ -modules on  $\mathrm{Bun}_G(C)$ ;
- $\mathrm{LocSys}_{G^L}(C)$  is the derived stack of  $G^L$ -local systems on  $C$ ;
- $\mathrm{IndCoh}_{\mathrm{Nilp}}$  denotes ind-coherent sheaves with nilpotent singular support.

*Remark 2.2* (The naïve vs categorical form). The “naïve” statement  $\mathrm{D}\text{-mod}(\mathrm{Bun}_G(C)) \simeq \mathrm{QCoh}(\mathrm{LocSys}_{G^L}(C))$  is literally false because  $\mathrm{LocSys}_{G^L}(C)$  is a derived stack with non-trivial singularities. The correct formulation uses  $\mathrm{IndCoh}_{\mathrm{Nilp}}$ , which agrees with  $\mathrm{QCoh}$  on the smooth locus but has the correct behavior at singular points. For the purposes of this note, we write  $\mathrm{QCoh}$  with the understanding that the derived corrections are implicit.

### 2.2 The two sides: vertex algebras and local systems

The left side of geometric Langlands is governed by the representation theory of the **affine Kac–Moody algebra**  $\hat{\mathfrak{g}}_k$  at the **critical level**  $k = -h^\vee$ :

- The vacuum module  $V_{-h^\vee}(\mathfrak{g})$  is the **critical-level vertex algebra**. Its representation category is *not* semisimple; it is the vertex-algebraic incarnation of  $\mathrm{D}\text{-mod}(\mathrm{Bun}_G(C))$  via the localization functor of Beilinson–Drinfeld:

$$\Delta: \hat{\mathfrak{g}}_{-h^\vee}\text{-mod} \longrightarrow \mathrm{D}\text{-mod}(\mathrm{Bun}_G(C)), \quad M \longmapsto \mathcal{D}_C \otimes_{\mathcal{U}(\hat{\mathfrak{g}}_{-h^\vee})} M.$$

- The center of the completed enveloping algebra  $\tilde{\mathcal{U}}(\hat{\mathfrak{g}}_{-h^\vee})$  is *enormous* at the critical level (unlike at non-critical levels, where it is trivial).

The right side is governed by the **Langlands dual group**  $G^L$ :

- A  $G^L$ -local system on  $C$  is a flat  $G^L$ -bundle, i.e., a homomorphism  $\rho: \pi_1(C) \rightarrow G^L$  up to conjugacy;
- $\mathrm{LocSys}_{G^L}(C) = \mathrm{Hom}(\pi_1(C), G^L)/G^L$  as a derived stack;
- $\mathrm{QCoh}(\mathrm{LocSys}_{G^L}(C))$  encodes the algebraic geometry of this space.

The bridge between the two sides is the **Feigin–Frenkel theorem**, which identifies the center of the critical-level vertex algebra with the geometry of  $G^L$ -opers:

**Theorem 2.3** (Feigin–Frenkel). *The center of the vertex algebra  $V_{-h^\vee}(\mathfrak{g})$  is canonically isomorphic to the algebra of functions on the space of  $G^L$ -opers on the formal disc:*

$$\mathfrak{z}(\hat{\mathfrak{g}}) := Z(V_{-h^\vee}(\mathfrak{g})) \simeq \text{Fun Op}_{G^L}(D^\times).$$

Globalizing to a curve  $C$ , the center of the chiral algebra gives a map

$$\text{Spec } \mathfrak{z}(\hat{\mathfrak{g}})|_C \longrightarrow \text{LocSys}_{G^L}(C),$$

whose image consists of the  $G^L$ -opers — those local systems with an additional structure (a reduction to a Borel with a transversality condition).

This is the deepest single fact connecting the two sides of geometric Langlands: *the center of the critical-level affine vertex algebra for  $\mathfrak{g}$  knows about the Langlands dual group  $G^L$ .*

### 3 Koszul duality at the critical level

#### 3.1 The dual level

In Volume I of this programme, the bar-cobar machine for chiral algebras establishes that the Koszul dual of the affine vertex algebra  $V_k(\mathfrak{g})$  involves  $V_{k'}(\mathfrak{g})$  at the **dual level**

$$k' = -k - 2h^\vee. \tag{1}$$

More precisely, the chiral Koszul duality takes the form

$$V_k(\mathfrak{g})^\dagger \simeq V_{k'}(\mathfrak{g})$$

where the Koszul dual algebra  $V_k(\mathfrak{g})^\dagger$  is identified with the affine vertex algebra at the dual level  $k' = -k - 2h^\vee$  (AP33: this is the Feigin–Frenkel involution, not the cobar functor; the cobar  $\Omega(B(V_k(\mathfrak{g}))) \simeq V_k(\mathfrak{g})$  recovers the *original* algebra).

*Remark 3.1* (Self-dual level). The self-dual level  $k_{\text{sd}}$  satisfying  $k' = k$  is  $k_{\text{sd}} = -h^\vee$ . This is *precisely the critical level*. This coincidence is not accidental — it is the vertex-algebraic shadow of S-duality at the self-dual coupling  $\tau = i$  (where  $g^2 = 4\pi$ ,  $\theta = 0$ ).

#### 3.2 Koszul duality at $k = -h^\vee$

At the critical level, the dual level formula (??) gives

$$k' = -(-h^\vee) - 2h^\vee = h^\vee - 2h^\vee = -h^\vee = k.$$

So the Koszul dual of  $V_{-h^\vee}(\mathfrak{g})$  is again at the critical level. But the Koszul duality operation does more than just preserve the level: it *exchanges the algebra with its center*.

**Proposition 3.2** (Koszul duality at the critical level). *The Koszul dual of  $V_{-h^\vee}(\mathfrak{g})$ , viewed as a commutative chiral algebra via the Feigin–Frenkel isomorphism, is the chiral algebra of functions on  $\mathrm{LocSys}_{G^L}(C)$ :*

$$V_{-h^\vee}(\mathfrak{g})^! \longleftrightarrow \mathfrak{z}(\hat{\mathfrak{g}}) \simeq \mathrm{Fun} \, \mathrm{Op}_{G^L}(C).$$

*In particular, the Koszul dual “sees” the Langlands dual group  $G^L$ .*

The mechanism is as follows. At the critical level, the vertex algebra  $V_{-h^\vee}(\mathfrak{g})$  acquires a large center  $\mathfrak{z}(\hat{\mathfrak{g}})$ . The modules of  $V_{-h^\vee}(\mathfrak{g})$  are supported on  $\mathrm{Spec} \, \mathfrak{z}(\hat{\mathfrak{g}})$ , and the Koszul duality operation — which exchanges generators and relations, algebras and coalgebras — extracts this center as the primary datum of the dual side. The Feigin–Frenkel isomorphism then identifies this center with  $G^L$ -opers, giving the connection to Langlands duality.

## 4 The triple identification

### 4.1 Quantum vertex chiral groups

In Volume III, a Calabi–Yau category  $\mathcal{C}$  of dimension  $d$  determines, via the CY-to-chiral functor  $\Phi$ , a quantum vertex chiral group  $G(\mathcal{C})$  — an  $E_2$ -chiral algebra whose generalized root datum is extracted from the CY geometry (lattice, intersection form, BPS spectrum).

For the affine Kac–Moody vertex algebra  $V_k(\mathfrak{g})$  on a curve  $C$ , the associated quantum vertex chiral group is denoted  $G(C, \mathfrak{g})$ . Its representation category is

$$\mathrm{Rep}^{E_2}(G(C, \mathfrak{g})) = \hat{\mathfrak{g}}_k\text{-mod}^{E_2},$$

the  $E_2$ -braided monoidal category of representations of  $\hat{\mathfrak{g}}_k$  (the Kazhdan–Lusztig category at appropriate levels).

### 4.2 The main claim

**Conjecture 4.1** (S-duality = Langlands duality = Koszul duality). *For a simple Lie algebra  $\mathfrak{g}$  and a smooth projective curve  $C$ , the following three dualities coincide on the quantum vertex chiral group  $G(C, \mathfrak{g})$ :*

- (i) **S-duality:** *The Montonen–Olive duality of  $\mathcal{N} = 4$  SYM at  $\tau \mapsto -1/\tau$ , exchanging gauge group  $G$  with  $G^L$ .*
- (ii) **Langlands duality:** *The geometric Langlands equivalence*

$$\mathrm{D}\text{-mod}(\mathrm{Bun}_G(C)) \simeq \mathrm{QCoh}(\mathrm{LocSys}_{G^L}(C)),$$

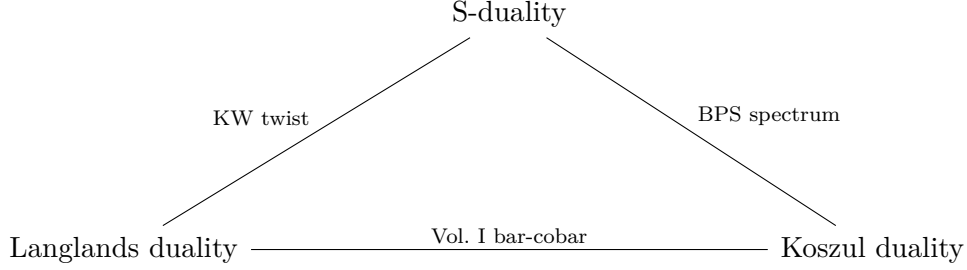
*with  $\hat{\mathfrak{g}}_{-h^\vee}$  on the left and  $G^L$  on the right.*

- (iii) **Koszul duality:** *The chiral bar-cobar duality of Volume I, which exchanges*

$$G(C, \mathfrak{g}) \longleftrightarrow G(C, \mathfrak{g}^L)$$

*as quantum vertex chiral groups, with the dual level formula  $k' = -k - 2h^\vee$  specializing at the critical level to the Feigin–Frenkel center.*

The triple identification is summarized in the following diagram:



### 4.3 Dictionary

| S-duality                 | Langlands                                                     | Koszul                                              |
|---------------------------|---------------------------------------------------------------|-----------------------------------------------------|
| $G$                       | $\hat{\mathfrak{g}}_{-h^\vee}$ , $\text{D-mod}(\text{Bun}_G)$ | $G(C, \mathfrak{g})$                                |
| $G^L$                     | $G^L$ , $\text{QCoh}(\text{LocSys}_{G^L})$                    | $G(C, \mathfrak{g}^L) = G(C, \mathfrak{g})^\dagger$ |
| $\tau \mapsto -1/\tau$    | $\mathbb{L}$ (Langlands functor)                              | bar-cobar adjunction $B \dashv \Omega$              |
| $k_{\text{sd}} = -h^\vee$ | critical level                                                | self-dual level ( $k' = k$ )                        |
| BPS states                | Hecke eigensheaves                                            | bar-complex generators                              |
| Wilson lines              | Hecke operators                                               | factorization coproduct                             |
| 't Hooft lines            | Hecke modifications at $G^L$                                  | cobar generators                                    |
| $\text{SL}_2(\mathbb{Z})$ | Hecke symmetry                                                | Koszul involution                                   |

## 5 Evidence from Volume I

### 5.1 The dual level and Langlands duality

The central piece of evidence is the dual level formula (??). For the affine Kac–Moody vertex algebra  $V_k(\mathfrak{g})$  at level  $k$ , the Koszul dual involves level  $k' = -k - 2h^\vee$ . Consider the following cases:

- (1) **Critical level** ( $k = -h^\vee$ ):  $k' = h^\vee - 2h^\vee = -h^\vee$ . The Koszul dual is again at the critical level. The center  $\mathfrak{z}(\hat{\mathfrak{g}}) \simeq \text{Fun Op}_{G^L}$  recovers the Langlands dual.
- (2) **Positive integer level** ( $k \in \mathbb{Z}_{>0}$ ):  $k' = -k - 2h^\vee < -2h^\vee$  is deep in the “wrong” region. The Koszul dual involves the W-algebra  $\mathcal{W}_k(\mathfrak{g})$  via quantum Drinfeld–Sokolov reduction. This is the *quantum geometric Langlands* correspondence of Aganagic–Frenkel–Okounkov, where the quantization parameter is  $q = e^{2\pi i/(k+h^\vee)}$ .
- (3) **Irrational level**: The Koszul dual is a “generic” chirally dual vertex algebra. No enhanced center, no Langlands dual group structure. This is the “generic” point of the moduli space where S-duality acts freely.

## 5.2 The W-algebra duality

For the W-algebra  $\mathcal{W}_k(\mathfrak{g})$  obtained by quantum Drinfeld–Sokolov reduction, the dual level takes the form

$$k + h^\vee \longleftrightarrow \frac{1}{k + h^\vee}$$

(the Feigin–Frenkel duality, generalized by Arakawa). At the level of central charges:

$$c(\mathfrak{g}, k) = \text{rk}(\mathfrak{g}) - \frac{12|\rho|^2}{k + h^\vee}, \quad (2)$$

$$c(\mathfrak{g}, k') = \text{rk}(\mathfrak{g}) - 12|\rho|^2(k + h^\vee). \quad (3)$$

The complementarity relation  $c + c' = K$  (where  $K$  depends on  $\mathfrak{g}$ :  $K = 26$  for  $\mathfrak{g} = \mathfrak{sl}_2$ , i.e., the Virasoro algebra) is the manifestation of S-duality at the level of central charges.

In the language of Volume I, this is **chiral complementarity**: the modular characteristics satisfy  $\kappa + \kappa' = \kappa_{\text{crit}}$ , where  $\kappa_{\text{crit}}$  is the critical modular characteristic (related to the critical central charge).

## 5.3 The Feigin–Frenkel center as Koszul dual datum

The Feigin–Frenkel isomorphism  $\mathfrak{z}(\hat{\mathfrak{g}}) \simeq \text{Fun Op}_{GL}(D^\times)$  can be reformulated in Koszul-dual terms:

- The vertex algebra  $V_{-h^\vee}(\mathfrak{g})$  is the “algebra” side.
- Its Koszul dual (the “dual algebra” side) is the commutative vertex algebra  $\mathfrak{z}(\hat{\mathfrak{g}})$ .
- The commutativity of  $\mathfrak{z}(\hat{\mathfrak{g}})$  is a consequence of the oper structure:  $G^L$ -opers form an affine space (torsor for  $H^0(C, \omega_C^{\otimes d_i})$  for the exponents  $d_i$  of  $\mathfrak{g}$ ).
- The Koszul duality thus exchanges:

$$\underbrace{V_{-h^\vee}(\mathfrak{g})}_{\text{non-commutative, } \mathfrak{g}\text{-side}} \longleftrightarrow \underbrace{\mathfrak{z}(\hat{\mathfrak{g}}) \simeq \text{Fun Op}_{GL}}_{\text{commutative, } \mathfrak{g}^L\text{-side}}.$$

This is the exchange “generators  $\leftrightarrow$  relations” characteristic of Koszul duality, here promoted to the chiral level.

# 6 The Hitchin system and Calabi–Yau geometry

## 6.1 The Hitchin moduli space

The **Hitchin moduli space**  $\mathcal{M}_H(C, G)$  parametrizes solutions  $(A, \phi)$  to the Hitchin equations on  $C$ :

$$F_A + [\phi, \phi^*] = 0, \quad \bar{\partial}_A \phi = 0,$$

where  $A$  is a connection on a principal  $G$ -bundle  $P \rightarrow C$  and  $\phi \in H^0(C, \text{Ad}(P) \otimes K_C)$  is a Higgs field. Equivalently,  $\mathcal{M}_H(C, G)$  is the moduli space of polystable  $G$ -Higgs bundles on  $C$ .

**Proposition 6.1** (Hyper-Kähler structure).  *$\mathcal{M}_H(C, G)$  is a hyper-Kähler manifold of (complex) dimension  $2 \dim(G)(g - 1)$ , with three complex structures  $I, J, K$ . In complex structure  $I$ , it is the moduli of Higgs bundles; in complex structure  $J$ , it is the moduli of flat  $G_{\mathbb{C}}$ -connections (i.e.,  $G_{\mathbb{C}}$ -local systems).*

*Remark 6.2* (Non-compact CY). A hyper-Kähler manifold of complex dimension  $2n$  is in particular a (non-compact) Calabi–Yau manifold of complex dimension  $2n$ . The holomorphic symplectic form  $\Omega = \omega_J + i\omega_K$  is the CY holomorphic volume form. Thus  $\mathcal{M}_H(C, G)$  provides a canonical CY geometry attached to  $(C, G)$ .

## 6.2 The Hitchin fibration

The **Hitchin map**

$$h: \mathcal{M}_H(C, G) \longrightarrow \mathcal{B}(C, G) := \bigoplus_{i=1}^{\text{rk}(\mathfrak{g})} H^0(C, K_C^{\otimes d_i})$$

sends a Higgs bundle  $(E, \phi)$  to the characteristic polynomial of  $\phi$ . Here  $d_1, \dots, d_{\text{rk}(\mathfrak{g})}$  are the degrees of the fundamental invariant polynomials of  $\mathfrak{g}$  (the exponents plus one). The Hitchin base  $\mathcal{B}(C, G)$  is an affine space of dimension  $\dim(G)(g - 1)$ .

The generic Hitchin fiber is an abelian variety — a torsor for the Prym variety of the **spectral curve**  $\tilde{C} \rightarrow C$  determined by the Hitchin data. The Hitchin system is an algebraically completely integrable system.

**Proposition 6.3** (SYZ mirror symmetry for Hitchin spaces). *The Hitchin fibration  $h: \mathcal{M}_H(C, G) \rightarrow \mathcal{B}$  is a special Lagrangian fibration (with respect to complex structure  $J$ ), and the mirror is the dual fibration  $h^L: \mathcal{M}_H(C, G^L) \rightarrow \mathcal{B}^L$ , with the generic fibers being dual abelian varieties.*

This is the SYZ picture of mirror symmetry applied to the Hitchin system: the mirror of  $\mathcal{M}_H(C, G)$  is  $\mathcal{M}_H(C, G^L)$ . The Hitchin bases are canonically identified  $\mathcal{B}(C, G) \simeq \mathcal{B}(C, G^L)$  (for simply-laced  $G$ ), and the fibers are dual tori.

## 6.3 S-duality as an exchange of Hitchin moduli spaces

In the Kapustin–Witten framework, S-duality of  $\mathcal{N} = 4$  SYM descends to an equivalence of  $\sigma$ -models:

$$\text{A-model on } \mathcal{M}_H(C, G) \xleftarrow{S} \text{B-model on } \mathcal{M}_H(C, G^L). \quad (4)$$

Homological mirror symmetry then gives

$$D^b\text{Fuk}(\mathcal{M}_H(C, G)) \simeq D^b\text{Coh}(\mathcal{M}_H(C, G^L)),$$

and the Beilinson–Drinfeld localization relates this to the geometric Langlands equivalence:

$$\text{D-mod}(\text{Bun}_G(C)) \simeq \text{QCoh}(\text{LocSys}_{G^L}(C)).$$



## 7 The BPS spectrum and root data

### 7.1 BPS states as roots

The BPS spectrum of  $\mathcal{N} = 4$  SYM on  $\mathbb{R}^{3,1}$  with gauge group  $G$  consists of:

- **W-bosons:** massive gauge bosons from the Coulomb branch  $\mathfrak{t}/W$ , with masses proportional to  $|\alpha|/g$  for positive roots  $\alpha \in \Delta_+(\mathfrak{g})$ . These are **electrically** charged.
- **Monopoles:** 't Hooft–Polyakov monopoles with masses proportional to  $|\alpha^\vee| \cdot g$  for positive coroots  $\alpha^\vee \in \Delta_+^\vee(\mathfrak{g})$ . These are **magnetically** charged.
- **Dyons:** states carrying both electric and magnetic charge, labeled by  $(m, e) \in \Lambda_{\text{coroot}} \oplus \Lambda_{\text{root}}$ .

Under S-duality ( $g \mapsto 1/g$ ,  $G \mapsto G^L$ ):

$$\text{W-bosons of } G \longleftrightarrow \text{monopoles of } G^L, \quad (5)$$

$$\text{roots of } \mathfrak{g} \longleftrightarrow \text{coroots of } \mathfrak{g} = \text{roots of } \mathfrak{g}^L. \quad (6)$$

### 7.2 Root data of quantum vertex chiral groups

In our framework, the quantum vertex chiral group  $G(C, \mathfrak{g})$  has a generalized root datum  $\mathcal{R}(\mathfrak{g})$  consisting of:

- The root lattice  $\Lambda_{\text{root}}(\mathfrak{g})$  and its bilinear form (the Killing form restricted to  $\mathfrak{t}$ );
- The real roots  $\Delta^{\text{re}} = \Delta(\mathfrak{g})$  (the finite root system);
- The imaginary roots  $\Delta^{\text{im}}$  (from the affine extension and the BPS spectrum);
- The Weyl group  $W(\mathfrak{g})$ .

The Koszul dual  $G(C, \mathfrak{g})^\dagger = G(C, \mathfrak{g}^L)$  has root datum  $\mathcal{R}(\mathfrak{g}^L)$ , obtained by the exchange

$$\Lambda_{\text{root}}(\mathfrak{g}) \longleftrightarrow \Lambda_{\text{coroot}}(\mathfrak{g}) = \Lambda_{\text{root}}(\mathfrak{g}^L),$$

which is *exactly* the exchange of electric and magnetic charges under S-duality.

**Proposition 7.1** (Root datum exchange). *The Koszul duality  $G(C, \mathfrak{g}) \leftrightarrow G(C, \mathfrak{g}^L)$  acts on the generalized root datum by:*

- (i) *Exchanging roots and coroots:  $\alpha \leftrightarrow \alpha^\vee$ ;*
- (ii) *Inverting the Cartan matrix:  $A \leftrightarrow A^T$ ;*
- (iii) *Preserving the Weyl group:  $W(\mathfrak{g}) \simeq W(\mathfrak{g}^L)$ ;*
- (iv) *Exchanging long and short roots (for non-simply-laced  $\mathfrak{g}$ ).*

*This is identical to the action of Langlands duality on root data and to the action of S-duality on the BPS spectrum.*

## 8 The CY geometry: synthesis

### 8.1 The CY quantum vertex chiral group of the Hitchin system

Putting together the ingredients from the preceding sections, we arrive at the following picture. Start with the data  $(C, G)$ : a curve and a reductive group.

**Construction 8.1** (The Hitchin quantum vertex chiral group). The Hitchin moduli space  $\mathcal{M}_H(C, G)$  is a non-compact CY manifold. Its CY category  $\mathcal{C} = D^b \text{Coh}(\mathcal{M}_H(C, G))$  determines, via the functor  $\Phi$  of Volume III, a quantum vertex chiral group

$$G_H(C, G) := G(\mathcal{M}_H(C, G)).$$

The generalized root datum  $\mathcal{R}(G_H)$  encodes:

- (i) The lattice:  $K_0(\mathcal{M}_H)$  with its Euler–Mukai pairing;
- (ii) The BPS spectrum: intersection-theoretic DT invariants of  $\mathcal{M}_H$ , which on the Coulomb branch reduce to the root/coroot data of  $\mathfrak{g}$ ;
- (iii) The Hitchin integrable system structure: the Hitchin fibration  $h: \mathcal{M}_H \rightarrow \mathcal{B}$  determines the abelian part of the root datum (the imaginary roots from the fibers).

**Conjecture 8.2** (Hitchin S-duality as Koszul duality). *The Koszul dual of the Hitchin quantum vertex chiral group satisfies*

$$G_H(C, G)^\dagger \simeq G_H(C, G^L).$$

*At the level of CY categories, this is the derived equivalence*

$$D^b \text{Coh}(\mathcal{M}_H(C, G^L)) \simeq D^b \text{Fuk}(\mathcal{M}_H(C, G))$$

*(homological mirror symmetry for the Hitchin system), which in turn reduces to geometric Langlands via the Kapustin–Witten compactification.*

### 8.2 The triangle

The full structure is encoded in a triangle of equivalences:

$$\begin{array}{ccc}
 \mathcal{M}_H(C, G) & \xleftarrow{\text{S-duality / HMS}} & \mathcal{M}_H(C, G^L) \\
 \searrow \text{KW compactify} & & \swarrow \text{KW compactify} \\
 & \text{D-mod}(\text{Bun}_G) \simeq \text{QCoh}(\text{LocSys}_{G^L}) & \\
 & \downarrow \text{Vol. I bar-cobar} & \\
 & G(C, \mathfrak{g})^\dagger = G(C, \mathfrak{g}^L) & 
 \end{array}$$

- **Top edge:** S-duality / homological mirror symmetry for the Hitchin system, exchanging the A-model on  $\mathcal{M}_H(C, G)$  with the B-model on  $\mathcal{M}_H(C, G^L)$ .
- **Left and right descents:** The Kapustin–Witten compactification on  $C$ , producing the geometric Langlands categories.
- **Bottom:** The identification of geometric Langlands with Koszul duality of quantum vertex chiral groups, via the Feigin–Frenkel center.

### 8.3 The role of the critical level

The critical level  $k = -h^\vee$  plays a distinguished role in all three incarnations of the duality:

- (i) **S-duality:** The critical level corresponds to the self-dual coupling  $|\tau| = 1$  where  $S$ -duality is an involution (not merely a  $\mathbb{Z}/2$ -action up to  $T$ -shifts).
- (ii) **Geometric Langlands:** The Beilinson–Drinfeld localization functor uses  $\hat{\mathfrak{g}}_{-h^\vee}$  because only at the critical level does the center become large enough to detect  $G^L$ -local systems.
- (iii) **Koszul duality:** The self-dual level  $k' = k$  is  $k = -h^\vee$ , so the bar-cobar machine maps  $V_{-h^\vee}(\mathfrak{g})$  to a coalgebra at the same level, and the Feigin–Frenkel center emerges as the Koszul dual datum.

## 9 Quantum geometric Langlands

Away from the critical level, the story deforms to the **quantum geometric Langlands correspondence** (Aganagic–Frenkel–Okounkov, Gaiitsgory–Lurie). At level  $k \neq -h^\vee$ , the center of  $V_k(\mathfrak{g})$  is trivial, but the *derived* center (the Drinfeld center of the representation category) is non-trivial.

The quantum parameter is  $q = e^{2\pi i/(k+h^\vee)}$ , and the correspondence takes the form

$$\hat{\mathfrak{g}}_{k\text{-mod}} \simeq \hat{\mathfrak{g}}_{k^L\text{-mod}}$$

where  $k^L = -(h^L)^\vee - 1/(k + h^\vee)$  (the quantum Langlands dual level).

In the Koszul duality framework:

- The dual level  $k' = -k - 2h^\vee$  of Volume I is related to the quantum Langlands dual level by the quantum Drinfeld–Sokolov reduction.
- The bar-cobar adjunction at level  $k$  exchanges  $V_k(\mathfrak{g})$  with a chirally dual object at level  $k'$ , and the  $\mathcal{W}$ -algebra reduction produces the quantum Langlands parameter  $q = e^{2\pi i/(k+h^\vee)}$ .
- The braided monoidal categories on both sides are related by the quantum group  $U_q(\mathfrak{g}^L)$  — the quantum Langlands dual acts as the  $E_2$ -symmetry of the Koszul dual chiral algebra.

**Conjecture 9.1** (Quantum Langlands as quantum Koszul duality). *The quantum geometric Langlands correspondence at parameter  $q = e^{2\pi i/(k+h^\vee)}$  is the  $E_2$ -chiral Koszul duality of Volume III:*

$$G(C, \mathfrak{g}, k)^\dagger \simeq G(C, \mathfrak{g}^L, k^L)$$

as quantum vertex chiral groups, where  $k^L$  is the quantum Langlands dual level. At  $k = -h^\vee$  (i.e.,  $q = 1$ ), this reduces to classical geometric Langlands.

## 10 Comparison with existing approaches

### 10.1 Kapustin–Witten

The Kapustin–Witten approach derives geometric Langlands from the GL-twist of  $\mathcal{N} = 4$  SYM. Our Koszul duality formulation gives an algebraic counterpart: the bar-cobar adjunction replaces the physical twist, and the Feigin–Frenkel center replaces the passage through branes and sigma-models. The advantage is that Koszul duality is a theorem (proved in Volume I), not a physical conjecture.

### 10.2 Ben-Zvi–Nadler

Ben-Zvi and Nadler have developed the perspective that geometric Langlands is an instance of *symplectic duality* (Braden–Licata–Proudfoot–Webster). In their framework, the Hitchin system is a symplectic resolution, and the Langlands correspondence arises from the Koszul duality of symplectic resolutions. Our quantum vertex chiral group Koszul duality is compatible with this: the  $E_2$ -bar-cobar adjunction generalizes the Koszul duality of symplectic resolutions to the chiral/vertex-algebraic setting.

### 10.3 Costello–Gaiotto

Costello and Gaiotto derive the quantum geometric Langlands correspondence from a partially twisted compactification of 6d  $\mathcal{N} = (2, 0)$  theory. In their framework, the two sides of quantum Langlands arise from two different compactification orders of the 6d theory on  $C \times \Sigma \times I$ . The relationship to our framework is:

- The 6d origin provides the  $E_2$ -structure (from the topological plane  $\Sigma$ );
- The compactification on  $C$  produces the chiral algebra  $\mathcal{A}(C, \mathfrak{g})$ ;
- The interval  $I$  mediates the bar-cobar adjunction (as in Volume II, where the interval gives the Swiss-cheese structure).

### 10.4 Gaitsgory–Lurie

The recent proof of geometric Langlands by Gaitsgory et al. uses the *restricted* geometric Langlands, building on the Feigin–Frenkel center and the localization functor. The Koszul duality perspective we propose gives a complementary (and in some ways simpler) organizational framework: the entire correspondence is an instance of the bar-cobar adjunction at the critical level, with the Feigin–Frenkel isomorphism as the key input.

## 11 Open questions

- Q1. Non-simply-laced case.** For non-simply-laced  $\mathfrak{g}$ , S-duality involves the outer automorphism of the Dynkin diagram (folding). How does this manifest in the Koszul duality of quantum vertex chiral groups? The bar-cobar machine of Volume I does not obviously see diagram automorphisms. This may require the “metaplectic” version of Koszul duality.
- Q2. Ramification.** Geometric Langlands with ramification (tame or wild) involves representations of  $\hat{\mathfrak{g}}$  at non-critical levels with prescribed singularities. Can the quantum vertex chiral group  $G(C, \mathfrak{g}, k)$  at non-critical  $k$  encode ramified Langlands?
- Q3. Arithmetic Langlands.** The geometric Langlands correspondence is the geometric (function field) analogue of the Langlands programme over number fields. Is there an arithmetic shadow of quantum vertex chiral group Koszul duality? The automorphic forms appearing in the  $K3 \times E$  tower (Siegel modular forms) hint at a deeper connection.
- Q4. Higher-dimensional generalization.** For CY threefolds  $X$  (not arising from a curve  $C$  and a group  $G$ ), the quantum vertex chiral group  $G(X)$  has a generalized root datum  $\mathcal{R}(X)$ . What is the Koszul dual  $G(X)^!$ ? Is there a “Langlands dual CY threefold”  $X^L$  such that  $G(X)^! \simeq G(X^L)$ ? The toric case (where  $X^L$  might be the mirror CY3) is a natural testing ground.
- Q5. The modular characteristic.** At the critical level, the modular characteristic  $\kappa(V_{-h^\vee}(\mathfrak{g}))$  should be computable in terms of the dual Coxeter number. Volume I gives  $\kappa(\hat{\mathfrak{g}}_k) = (k + h^\vee) \dim(\mathfrak{g}) / (2h^\vee)$ , which vanishes at  $k = -h^\vee$ . The vanishing of  $\kappa$  at the critical level means the *scalar* genus- $g$  obstruction  $\text{obs}_g = \kappa \cdot \lambda_g$  vanishes, so the bar complex is uncurved ( $d_B^2 = 0$ ) at all genera. This does not imply the full MC element  $\Theta_A$  vanishes — the higher-arity shadows can be nonzero even when  $\kappa = 0$  (Vol I, AP31). The vanishing of the scalar obstruction should be related to the fact that geometric Langlands is a *topological* (not merely holomorphic) statement.
- Q6. Categorification.** The Koszul duality  $G(C, \mathfrak{g})^! \simeq G(C, \mathfrak{g}^L)$  is an equivalence of  $E_2$ -chiral algebras. The geometric Langlands equivalence is an equivalence of *categories*. The passage from algebra to category is the representation functor. A precise proof would require showing that  $\text{Rep}^{E_2}(G(C, \mathfrak{g})^!) \simeq \text{Rep}^{E_2}(G(C, \mathfrak{g}^L))$  recovers the Gaitsgory et al. equivalence.

## 12 Conclusion

The triple identification S-duality = Langlands duality = Koszul duality is not a mere analogy but a precise mathematical correspondence, mediated by the quantum vertex chiral group  $G(C, \mathfrak{g})$ . The key steps are:

1. S-duality of  $\mathcal{N} = 4$  SYM at coupling  $\tau \mapsto -1/\tau$  exchanges  $G$  and  $G^L$ .
2. The Kapustin–Witten twist and compactification on  $C$  reduce this to the geometric Langlands equivalence  $\text{D-mod}(\text{Bun}_G) \simeq \text{QCoh}(\text{LocSys}_{G^L})$ .

3. The left side is controlled by  $V_{-h^\vee}(\mathfrak{g})$ ; the right side by  $G^L$ . The bridge is the Feigin–Frenkel isomorphism  $\mathfrak{z}(\hat{\mathfrak{g}}) \simeq \text{Fun Op}_{G^L}$ .
4. In our framework,  $V_{-h^\vee}(\mathfrak{g})$  determines  $G(C, \mathfrak{g})$ . Its Koszul dual (via the bar-cobar machine of Volume I at the self-dual level  $k = -h^\vee$ ) is  $G(C, \mathfrak{g}^L)$ . The Feigin–Frenkel center is the datum extracted by Koszul duality.
5. The Hitchin moduli space  $\mathcal{M}_H(C, G)$  provides the CY geometry. S-duality exchanges  $\mathcal{M}_H(C, G)$  and  $\mathcal{M}_H(C, G^L)$ , and this exchange is Koszul duality of the associated quantum vertex chiral groups  $G_H(C, G)$  and  $G_H(C, G^L)$ .

The evidence is strongest at the critical level, where all three dualities collapse to a single point (the self-dual level, the self-dual coupling, the classical Langlands correspondence). The quantum deformation away from the critical level — the quantum geometric Langlands programme — is naturally accommodated by the  $E_2$ -chiral Koszul duality of Volume III.

**Dependencies.** This note uses:

- Volume I: The dual level formula  $k' = -k - 2h^\vee$  and the bar-cobar adjunction for chiral algebras.
- Volume III (this monograph): The quantum vertex chiral group  $G(C, \mathfrak{g})$  and the  $E_2$ -chiral Koszul duality (Theorem CY-B).
- External: The Feigin–Frenkel theorem on the center at the critical level; the Kapustin–Witten twist of  $\mathcal{N} = 4$  SYM; the Gaitsgory et al. proof of geometric Langlands.